

Predicting Formula 1 Lap Times and Team Pit Windows

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Project question This project uses Formula 1 timing data to predict lap times and pit-stop windows. Our goal is to understand whether information about nearby cars and recent pace improves these predictions, and whether the same approach produces strong results with new cars or tracks.

1 Overview

In Formula 1, teams must decide when to stop, which tires to use and how to respond to nearby cars. As fans, we can observe timing and tire age, but we do not know the true fuel load, tire condition or which engine mode the team is using. We therefore rely on publicly available data to build our predictions rather than treating estimates of those hidden quantities as known inputs.

Our approach is to keep the model simple and add three observable predictors: *attacking*, *defending*, and *relative pace*. We first test these in a linear lap-time model, then separately use a random forest to predict when a team will make its actual first stop.



2024 Miami GP race start. Credit: McLaren.com.

2 Data and evaluation

The models use data from 2022–2025, during the [2022 technical regulations](#). We train and evaluate on 2022–2024, then test on 2025. Additional tests on selected 2026 races examine performance with new cars and circuits.

The data comes from [FastF1](#), using standard-format race weekends rather than sprint weekends. Within these weekends, we select dry, green-flag laps with no pit stops to focus on normal racing conditions. Yellow and red flag periods would be better handled in a full strategy simulator because they interrupt normal racing and "fix" lap times.

Cleaning step	Removed at step	Laps remaining
Raw race laps	—	78,297
Keep slick tires	4,773	73,524
Keep green-flag laps	6,707	66,817
Remove pit-in and pit-out laps	3,419	63,398
Require accurate timing	841	62,557
Other timing and opening-lap checks	0	62,557
Require known tire age	68	62,489

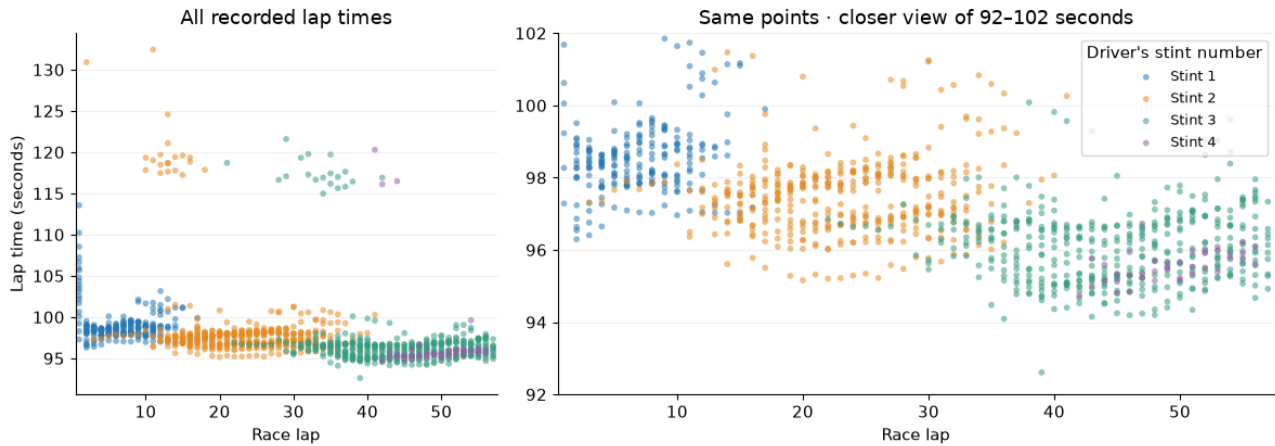
Table 1: Data cleaning process.

We shuffle the usable 2022–2024 laps and use 70% for training and 30% for evaluation, split equally between tuning and validation. We then refit on all development laps and test on 2025.

3 Limitations of timing data

When I started modeling this, my intuition was to predict lap time through tire degradation, but some timing patterns remain relatively constant and do not provide a clear degradation signal on their own. Observed lap and sector times combine fuel burn, tires, traffic and driving decisions, so improvements from lower fuel weight can offset tire degradation. This led me to add race-context predictors rather than trying to infer fuel consumption, tire degradation or engine modes from timing alone.

Bahrain 2024 · lap times combine fuel, tyre and race-context effects



We use Bahrain because there were no safety cars during the race.

Bahrain 2024 · sector and lap times outside pit stops



Average lap time fell by roughly two seconds during the race.

4 Adding race context

A first approach is to use tire age, tire compound, race progress, pre-race form and pack pace, since these observable quantities describe the conditions of a lap. Tire age and race progress provide useful timing information, but they do not directly measure tire wear or fuel load, whose relationships with pace can vary by track. We add nearby-car counts and recent relative pace to describe situations consistent with attacking, defending or managing pace, without assuming that these inputs reveal a driver's intentions.



2022 Monaco GP. Credit: F1i.com.

4.1 Scaling lap times

We need a common scale across tracks because different circuit lengths and layouts produce different typical lap times. We use a pre-race reference from qualifying, or practice when qualifying is not usable, so the reference is available before the race. Practice supplies the reference for four races in the reported results.

Let $\mathcal{L}_{i,t}$ be driver i 's lap time, and let $\mathcal{L}_{\text{ref}}$ be the median of drivers' best eligible laps in the chosen reference session. We predict a relative time and then convert it back to seconds:

$$y_{i,t} = \frac{\mathcal{L}_{i,t}}{\mathcal{L}_{\text{ref}}} - 1, \quad \widehat{\mathcal{L}}_{i,t} = \mathcal{L}_{\text{ref}}(1 + \widehat{y}_{i,t}).$$

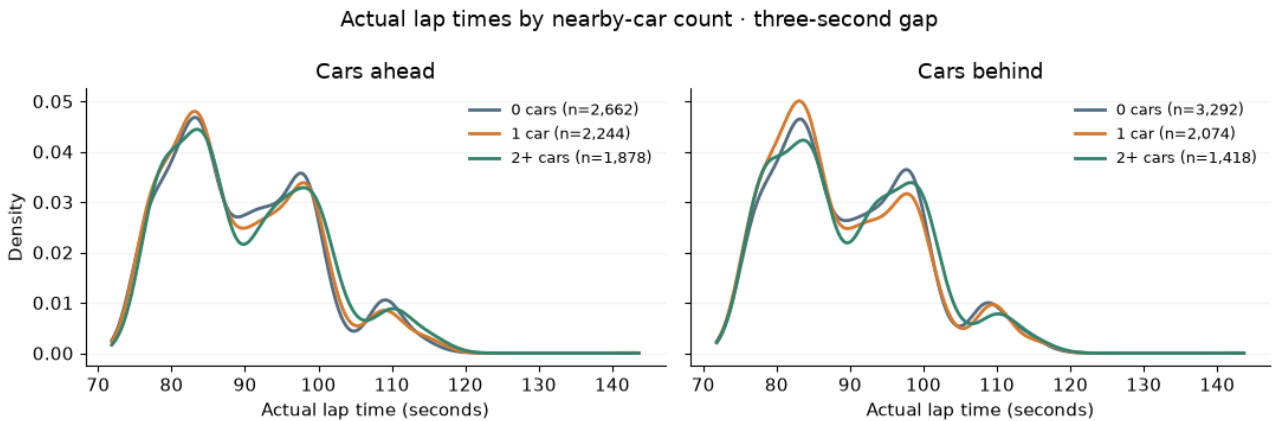
A one-second difference represents a larger share of a short lap than of a long lap, so dividing by the reference puts circuits on a comparable scale. Taking the median reduces the influence of an unusually quick or slow driver, although practice and qualifying conditions can differ.

4.2 Attacking and Defending

A driver close to another car may face different conditions from a driver in clear air, so we use the number of nearby cars as a simple description of traffic. We count cars within x_A seconds ahead for attacking and within x_D seconds behind for defending. We allow different thresholds because cars ahead and behind may have different relationships with lap time:

$$A_{i,t}(x_A) = \sum_{j \neq i} \mathbb{I}\{0 < g_{ij,t}^{\text{ahead}} \leq x_A\}, \quad D_{i,t}(x_D) = \sum_{j \neq i} \mathbb{I}\{0 < g_{ij,t}^{\text{behind}} \leq x_D\}.$$

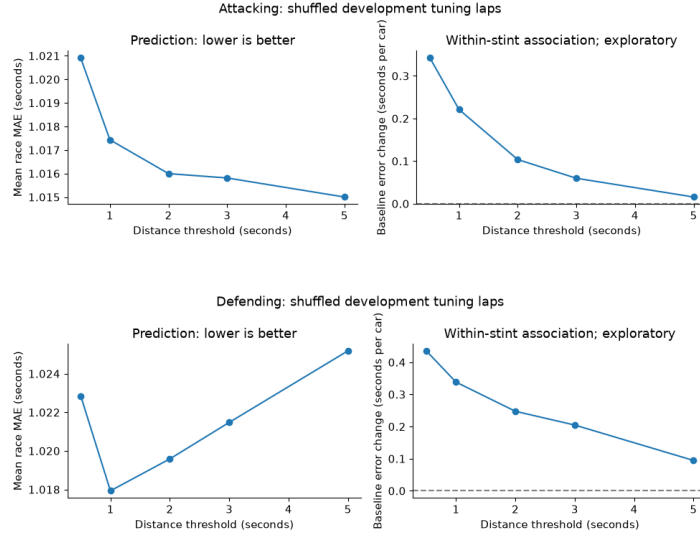
The gaps use completed timing observations available at lap start. These labels describe proximity and do not establish that an overtake or a defensive move occurred.



4.3 Learning the traffic thresholds

We want to know how close surrounding cars need to be before counting them helps predict lap time, so we compare $x \in \{0.5, 1, 2, 3, 5\}$ seconds on the tuning laps. We select the value with the lowest mean absolute error (MAE), which measures the average size of our prediction errors in seconds. This selects a threshold for prediction rather than choosing whichever produces the strongest apparent association. We select five seconds for attacking and one second for defending, then keep these choices fixed.

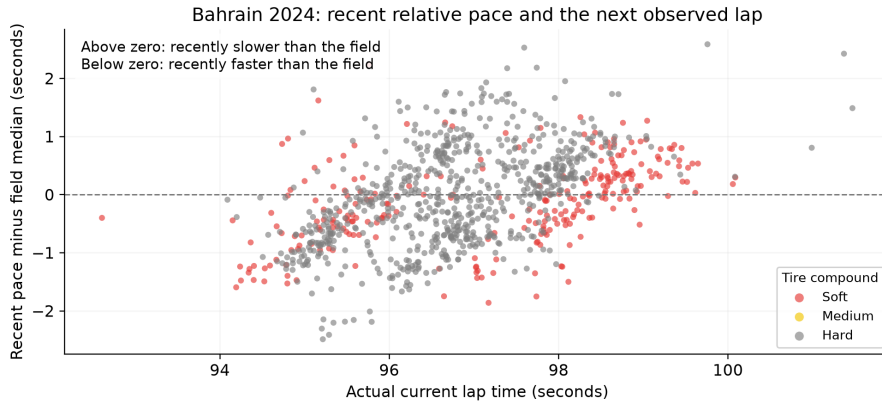
$$\hat{x} = \arg \min_x \frac{1}{R} \sum_{r=1}^R \text{MAE}_r(x).$$



4.4 Relative pace

We also want to know whether driving faster or slower than the surrounding field helps predict the next lap. We compare the mean of up to three eligible completed laps with the median of the available drivers' recent means, including the driver's own, and divide by the reference time:

$$P_{i,t} = \frac{\overline{\mathcal{L}}_{i,t}^{\text{recent}} - \text{median}_j(\overline{\mathcal{L}}_{j,t}^{\text{recent}})}{\mathcal{L}_{\text{ref}}}.$$



4.5 Circuit and weather information

Circuit and weather conditions also influence how teams prepare for a race. Mexico City’s high altitude and thin air, for example, affect downforce and cooling, motivating changes to the cars’ setup.

To capture some of these differences, we add seven circuit inputs: length, turns, altitude, elevation change, longest straight, average corner speed and a street-circuit indicator. We also add pre-race air and track temperature to describe the race conditions. These nine inputs are tested together, so this comparison does not separate the effects of circuit characteristics and weather.

For this project, the same circuit inputs are reused across seasons.



Carlos Sainz at the 2024 Mexico GP. Credit: RacingNews365.com.

5 Impact of the predictors

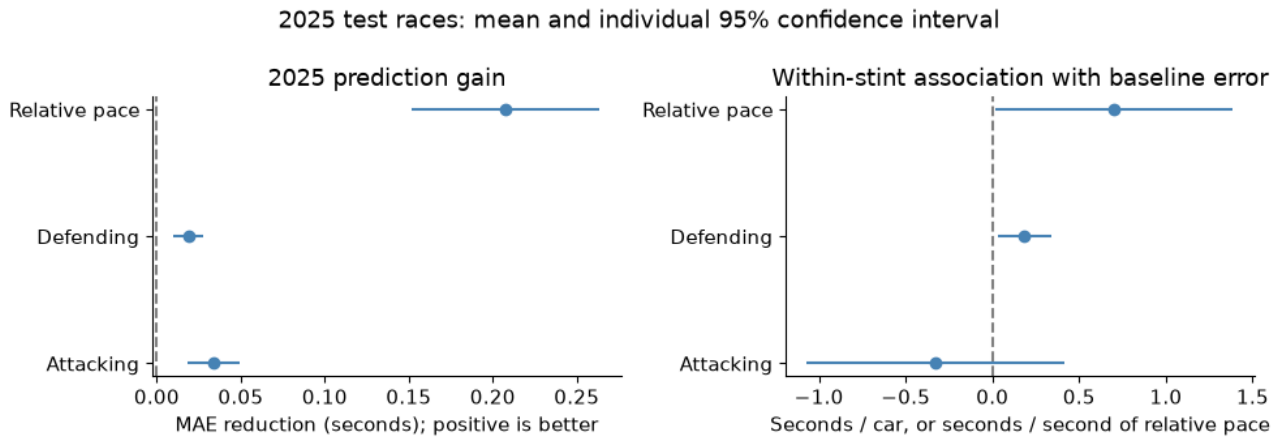
To assess each predictor’s contribution, we compare the basic model with an otherwise equivalent model containing one additional feature. We fit both on development data and subtract their MAEs within each 2025 race. A positive gain means that the extra input improved prediction:

$$\Delta_r = \text{MAE}_{r,\text{baseline}} - \text{MAE}_{r,\text{baseline+feature}}, \quad H_0 : \mathbb{E}[\Delta_r] = 0.$$

We test each predictor separately using a t -test on the gains across races.

Added predictor	MAE gain (s)	Individual 95% CI	Individual p
Attacking	0.03413	[0.01868, 0.04958]	0.00022
Defending	0.01893	[0.01007, 0.02779]	0.00031
Relative pace	0.20780	[0.15203, 0.26357]	< 0.0001

All three predictors reduce average error, with relative pace giving the largest improvement.



6 The linear lap-time model

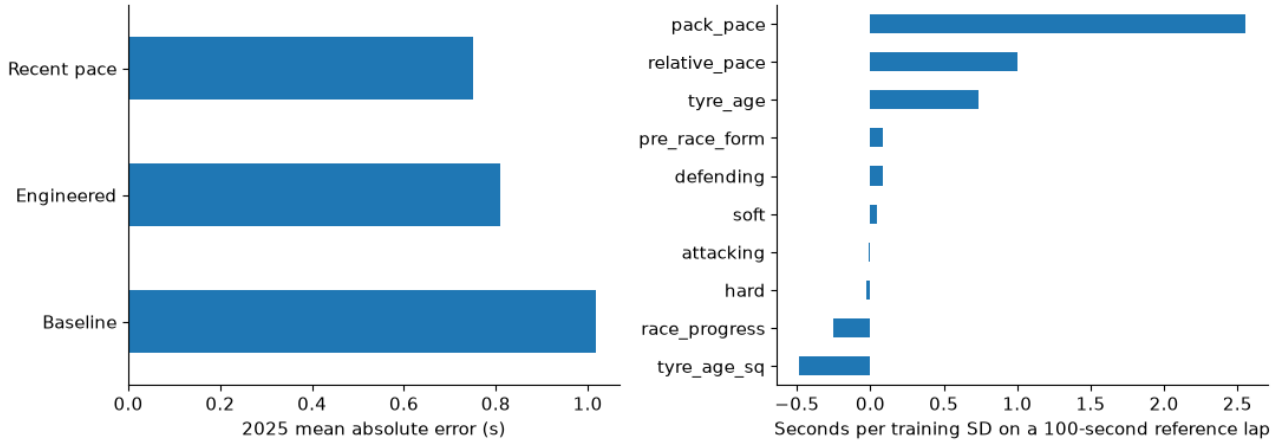
Interpretability is useful when working with new cars or circuits, so we start with ordinary least squares to combine the inputs and inspect their fitted contributions. Writing out the predictors, our model is:

$$\begin{aligned}\hat{y}_{i,t} = & \beta_0 + \beta_{\text{pack}} \widetilde{\text{Pack pace}}_{i,t} + \beta_{\text{form}} \widetilde{\text{Pre-race form}}_i \\ & + \beta_{\text{progress}} \widetilde{\text{Race progress}}_{i,t} + \beta_{\text{age}} \widetilde{\text{Tire age}}_{i,t} + \beta_{\text{age}^2} \widetilde{\text{Tire age squared}}_{i,t} \\ & + \beta_{\text{soft}} \mathbb{I}\{\text{Soft tire}\}_{i,t} + \beta_{\text{hard}} \mathbb{I}\{\text{Hard tire}\}_{i,t} \\ & + \beta_A \widetilde{\text{Attacking}}_{i,t} + \beta_D \widetilde{\text{Defending}}_{i,t} + \beta_P \widetilde{\text{Relative pace}}_{i,t}.\end{aligned}$$

Each β is a fitted coefficient, and the tilde indicates an input standardized using training data. Pack pace summarizes recent field pace, pre-race form describes the driver's reference-session pace, and race progress and tire age are expressed as fractions of the scheduled race distance. The soft- and hard-tire indicators compare those compounds with medium tires, while attacking, defending and relative pace are defined in Section 4.

We replace missing inputs with training medians.

Model	MAE (s)	95% CI for MAE	Within 1 second
Basic linear	1.018	[0.647, 1.388]	67.94%
Engineered linear	0.809	[0.422, 1.196]	79.06%
Engineered linear with weather and track data	0.741	[0.326, 1.156]	85.36%
Recent pace	0.750	[0.271, 1.228]	81.36%

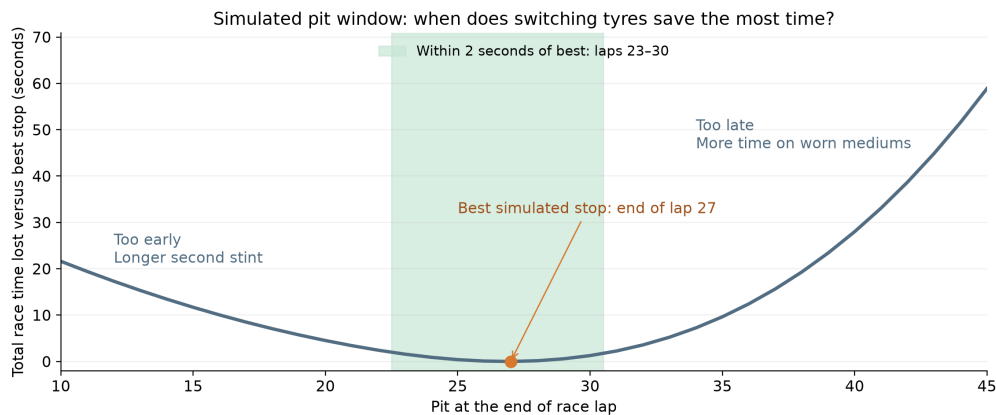


The table shows that adding race context reduces MAE by about 0.208 seconds compared with the basic linear model. We also predict each lap using only the driver's recent three-lap average, since a more elaborate model should be compared with information that is already easy to observe. This benchmark performs better at 0.750 seconds MAE, so we retain linear regression for its interpretability without claiming it is the most accurate predictor.

Adding circuit and weather data lowers 2025 MAE from 0.809 to 0.741 seconds, though the improvement is uncertain ($p = 0.196$). Validation MAE also falls from 0.751 to 0.737 seconds. We keep the model without these inputs as our simpler main example.

7 Predicting a team's pit window

In principle, a lap-time model could compare total race times for different pit laps and identify the fastest strategy, as the simulation below shows. Our current model cannot do this reliably because future traffic, tire wear and race interruptions are unknown. We therefore use a random forest to predict the team's actual pit window instead.



For efficiency, we make forecasts at laps 10, 20 and 30 for cars that have not yet stopped, giving us repeated decision points without creating a forecast at every lap. We use tire age, compound, recent pace, position, nearby cars and the fraction of rivals already observed stopping because these describe the situation available to the team at that time.

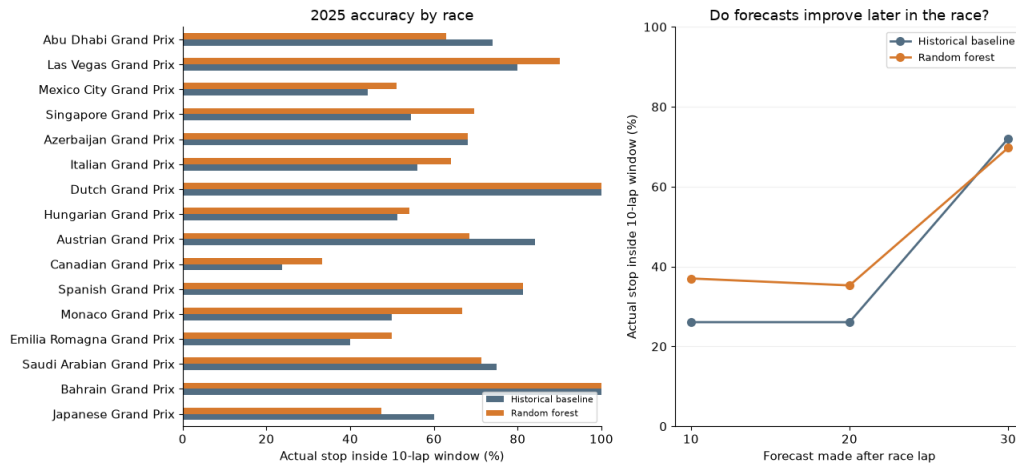
Predicting an exact stop lap would require more precision than we expect from public timing data, so we use a fixed ten-lap window as the target. The forest scores candidate windows and selects the highest-scoring one.

We use a random forest because a stop decision may depend on combinations of conditions, such as tire age together with traffic, that a simple additive model can miss. For this model, we shuffle and split whole development races 70/30, then use half the evaluation races to choose between two tree depths and half for validation.

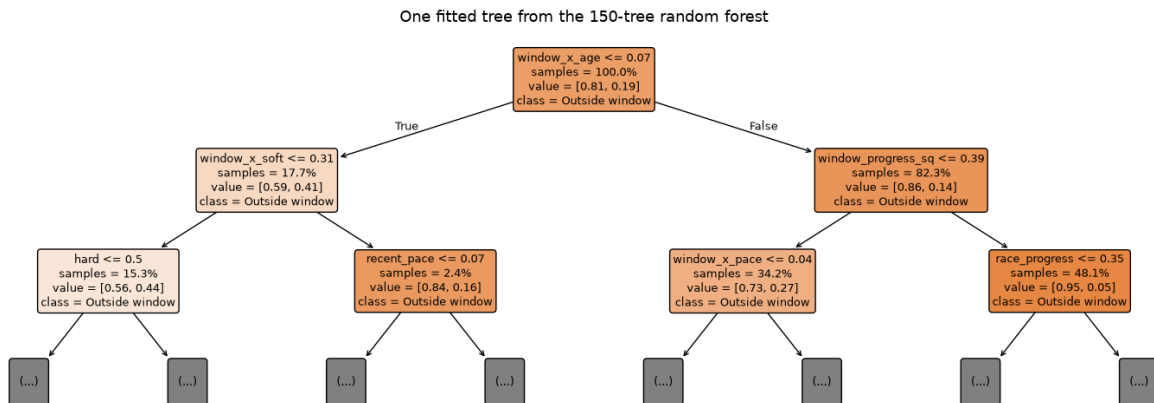
7.1 Pit-window results

The table below compares both methods on the same 356 forecasts across 16 races. We average accuracy within each race before averaging across races, so weekends with more eligible checkpoints do not receive extra weight.

Model	Window accuracy	Race-level 95% CI
Historical baseline	65.16%	[53.75%, 76.58%]
Random forest	67.42%	[57.51%, 77.32%]



The forest improves accuracy by 2.25 percentage points, but the difference is uncertain ($p = 0.362$). One tree is shown below:



8 Other models

In addition to the linear model, I trained a gradient boosting model and a small neural network to capture more flexible relationships between the inputs. For each method, we fit one version using race-context inputs and another adding the same circuit and weather data used in the linear extension.

We use the same development and 2025 evaluation samples for all four versions. The neural network has two hidden layers with 16 and 8 neurons, using ReLU activation.

Model	Circuit and weather data	MAE (s)	Within 1 second
Engineered linear (selected)	Without	0.809	79.06%
Gradient boosting	Without	0.725	82.32%
Gradient boosting	With	0.849	79.34%
Small neural network	Without	0.710	83.72%
Small neural network	With	0.853	82.97%

Both nonlinear models perform better than the selected linear model without the extra circuit and weather inputs. We retain linear regression as the main explanatory model for its interpretability, while recognizing that it does not give the lowest prediction error.

9 Conclusion

Adding traffic and relative pace reduces the linear model's MAE from 1.018 to 0.809 seconds. Relative pace gives the largest improvement. Circuit and weather data lower the error further to 0.741 seconds. The neural network gives the lowest error, but we keep the linear model as the main model because it is easier to explain.

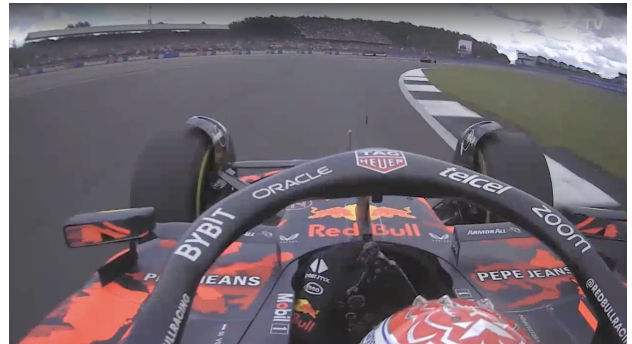
The random forest predicts the actual pit window with 67.42% accuracy, compared with 65.16% for the historical baseline. This is a small improvement, and it does not show that the predicted stop is the fastest strategy.

Future improvements I would like to add the following data:

- Fuel load and consumption, to separate the effect of a lighter car from tire wear.
- Tire condition and temperature, to estimate when changing tires could save time.
- Engine modes and brake temperatures, to better explain changes in pace.
- Onboard video, to compare racing lines, braking points and steering at the same point in a corner. The Silverstone images below show laps 46 and 16, but the different camera positions make a direct comparison difficult.
- Onboard audio, to check whether the engine sounds different at different points in the race, accounting for gear and speed.



Lap 46



Lap 16

A 2026 races test

These were the results:

Circuit	Lap-time model	Laps	MAE (s)	Within 1 second
Melbourne	Basic linear	814	1.171	48.03%
Melbourne	Engineered linear	814	0.721	77.76%
Melbourne	Engineered linear + track features	814	0.650	82.68%
Suzuka	Engineered linear	896	0.616	85.16%
Suzuka	Engineered linear + track features	896	0.630	83.59%
Madrid (new circuit)	Engineered linear	959	0.776	78.10%
Madrid (new circuit)	Engineered linear + track features	959	0.871	68.72%

At Madrid, a brand new circuit, the engineered model has an MAE of 0.776 seconds. Adding track features raises it to 0.871 seconds, so they do not improve this result.